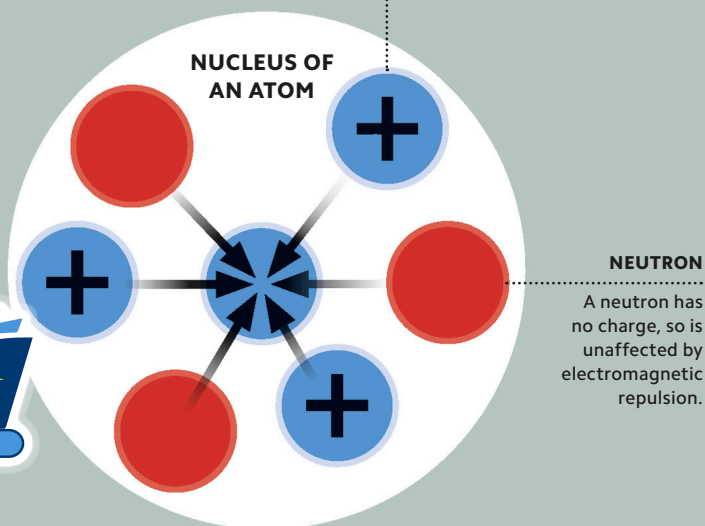


Nuclear force

The nuclear force binds quarks (and protons and neutrons) together, opposing electromagnetic repulsion (see p.22).

PROTON Protons repel each other, but this repulsion is overcome by the stronger nuclear force.



NEUTRON

A neutron has no charge, so is unaffected by electromagnetic repulsion.

HOLDING IT TOGETHER

Four fundamental forces are responsible for binding the matter particles in the universe together, and each is governed by quantum physics to some extent. The most powerful of these forces, known as the strong force, only works on tiny scales of about one million-billionth of a yard. This force bonds quark particles together to form protons and neutrons, and produces a nuclear force that binds these to form atomic nuclei.

The strong force is carried by particles called gluons.